★ Fuzzy set Theory

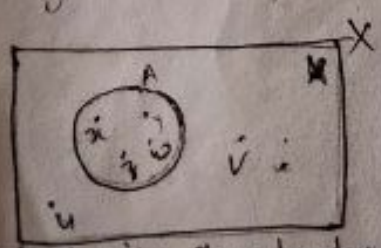
Introduction :⇒

Classical set theory (crisp set) provide a knowledge of ~~about~~ an element (member of set) in some universal set or universe of discourse. about their membership.

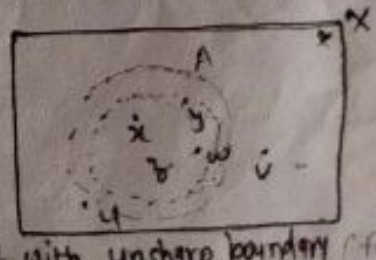
ie. whether it is member of a set or not a member of it. Hence, a sharp, unambiguous distinction exist betn the member and non member of the class represented by classical set. But many of the collections and categories we commanly use such as set of tall people, Expansive cars, highly contagious disease, number much greater than 1, or sunny day's etc. do not exhibit's this characteristics. Instead their boundaries seem vague. and the transition from member to non-member appears gradual rather than abrupt.

Thus to reduce the complexity, A fuzzy set theory was introduce by eliminating the sharp boundary in crisp set

A fuzzy set can be define by assigning a membership grade to each element in the universe of discourse. This grade correspond to the degree to which that ~~indivig~~ element is similar or compatible ~~to which that indiv~~ with the concept represented by the fuzzy set. Thus an element in universal set may belongs to fuzzy set to greater or lesser extent.



classical set with sharp boundary



Set with unsharp boundary (fuzzy set)

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Defⁿ : Crisp set / classical set

Let 'X' be the set of universe of discourse,

A set 'A' defined on element of 'X', such that

$$A = \{ x \mid x \in X \}$$

denoting that x belong to 'A' [or not belongs to 'A']

is called crisp set, if $x \in A$ has full membership in A

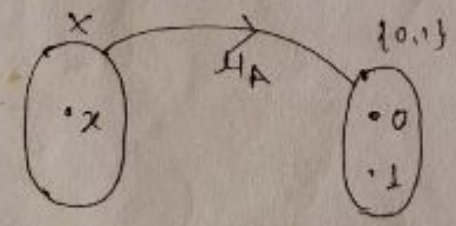
(i.e completely belongs to set 'A')

Here characteristic function (membership function) defined as

$$\mu_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$

i.e. if A is a crisp set defined on element of universal set X, then the characteristic function is mapping from X to set ~~of element~~ {0, 1}

$$\mu_A(x) : X \longrightarrow \{0,1\}$$



For eg: Let X (universal set) is the set of element

$$X = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \}$$

Then set $A = \{ x \mid 0 < x \leq 6 \}$

$= \{ 1, 2, 3, 4, 5, 6 \}$ is a crisp set

where,

$$\mu_A(x) = \begin{cases} 1, & \text{if } x = 1, 2, 3, 4, 5, 6 \\ 0, & \text{if } x = 0, 7, 8, 9 \end{cases}$$

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Defⁿ Fuzzy set:

It is a set with unsharp boundaries betn the member and nonmember of element x in X (i.e. set representing not only the membership or non membership of element in universal set X but also their partialy membership.)

Defⁿ: A set 'A' defined on elemement of universal set X. is said to be fuzzy set if their characteristic function (or membership function) is mapping from X to the interval [0,1]. (i.e. membership value of each element in A lies betn 0 to 1)

It is denoted and defined by representing all elements in x with their membership value (grade)

i.e. $$A = \{(x, \mu_A(x)) / \mu_A(x) \in [0,1], \text{ for all } x \in X\}$$



Representation of fuzzy set:

A fuzzy set 'A' defined on element $x_i \in X$, $i = 1,2,3 \ldots n$ is represented as follow

$$A = \{(x_1, \mu_A(x_1)), (x_2, \mu_A(x_2)) \ldots (x_n, \mu_A(x_n))\}$$

OR

$$A = \left\{ \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i} \right\}$$

$$= \left\{ \frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \ldots + \frac{\mu_A(x_n)}{x_n} \right\}$$

Note: This sign '+' does not represent the algebraic sum. It is only use to represent the distinct element with membership grade.

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Empty fuzzy set (ϕ):

A fuzzy set A is said to be empty fuzzy set iff membership value of all element $x \in X$ is zero in A

i.e. ϕ = Empty fuzzy set A

$$= \left\{ \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i} \;/\; \mu_A(x_i) = 0 \text{ for all } x_i \in X \right\}$$

Support of fuzzy set A : Supp(A)

It is a classical set A, of all element $x \in X$ for which $\mu_A(x) > 0$

i.e. $\text{Supp}(A) = \{ x \;/\; \mu_A(x) > 0 \text{ for all } x \in X \}$

For eg :⇒ let $X = \{a, b, c, d, e, f\}$ be the universal set and $A = \left\{ \frac{0.3}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0}{d} + \frac{0}{e} + \frac{0.8}{f} \right\}$ be the fuzzy set

Then,

$$\text{Supp}(A) = \{a, b, c, f\}$$

Nucleus of fuzzy set : N(A)

It is a classical set containing all element $x \in X$ for which $\mu_A(x) = 1$,

for ie $A = \left\{ \frac{0.7}{a} + \frac{0.3}{b} + \frac{1}{c} + \frac{0.1}{d} + \frac{1}{e} + \frac{0}{f} \right\}$

Then

$$N(A) = \{c, e\}$$

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★ α-level set (α-cut) :

It is a crisp set, in which for every element $x \in X$ $\mu_A(x) \geq \alpha$ where α is real number lies bet 0 & 1 and it is denoted by A_α or A^α or ${}^\alpha A$

★ strong α-cut :

It is a crisp set, in which for all $x \in X$. their membership grade $\mu_A(x) > \alpha$. when $\alpha \in [0,1]$ and it is denoted by $A_{\alpha+}$ or $A^{\alpha+}$ or ${}^{\alpha+}A$

For eg: let $A = \left\{ \frac{0.3}{x_1}, \frac{0.4}{x_2}, \frac{0.6}{x_3}, \frac{0.8}{x_4}, \frac{0.9}{x_5} \right\}$ be the fuzzy set then

$A_{0.4} = \{x_2, x_3, x_4, x_5\} = A_{0.3}^+$

$A_{0.8} = \{x_4, x_5\} = A_{0.6}^+$

$A_{0.9} = \{x_5\} = A_{0.8}^+$

★ Hight of fuzzy set

A Hight of a fuzzy set is the maximum membership grade attain by any element x in A fuzzy set A

For eg (i) If $A = \left\{ \frac{0.3}{x_1} + \frac{0.4}{x_2} + \frac{0.7}{x_3} + \frac{0.3}{x_4} \right\}$

$\Rightarrow$ hight of $A = 0.7$

(ii) If $A = \left\{ \frac{0.3}{x_1} + \frac{0.4}{x_2} + \frac{0.8}{x_3} + \frac{0.8}{x_4} \right\}$

$\Rightarrow$ Hight of $A = 0.8$

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Basic operations on fuzzy sets

(i) Union : let X be the universal set and let A & B are any two fuzzy set defined on element $x \in X$. with membership grade $\mu_A(x)$ & $\mu_B(x)$ respectively. Then their union is defined and denoted as

$$C = A \cup B = \left\{ \sum \frac{\mu_C(x)}{x} \;/\; \mu_C(x) = \max \cdot [\mu_A(x), \mu_B(x)] \text{ for all } x \in X \right\}$$

(ii) Intersection :

Let X be the universal set and A & B are any two fuzzy sets defined on all element $x \in X$ with membership grade $\mu_A(x)$ & $\mu_B(x)$ respectively

Then their intersection is given by

$$C = A \cap B = \left\{ \sum \frac{\mu_C(x)}{x} \;/\; \mu_C(x) = \min \cdot [\mu_A(x), \mu_B(x)] \text{ for all } x \in X \right\}$$

(iii) Complement :

let A be a fuzzy set defined on all element of universal set X. Then it's complement is denoted and define as. =

$$A^c = \overline{A} = \left\{ \sum \frac{\mu_{\overline{A}}(x)}{x} \;/\; \mu_{\overline{A}}(x) = [1 - \mu_A(x)], \text{ for all } x \in X \right\}$$

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(iv) Difference of two fuzzy set

Let X be the universal set, and A & B are any two fuzzy set defined for all $x \in X$. with membership grade $\mu_A(x)$ and $\mu_B(x)$ respectively. Then their difference is denoted and defined as

$$C = A \setminus B \left(\begin{matrix} OR \\ or \end{matrix} \quad \begin{matrix} A - B \\ A \cap \bar{B} \end{matrix} \right) = \left\{ \sum * \frac{\mu_C(x)}{x} \;/\; \mu_C(x) = \min \left[\mu_A(x), (1 - \mu_B(x)) \right] \right\}$$

OR

$$C = A \setminus B \text{ OR } A \cap \bar{B} = \left\{ \sum \frac{\mu_C(x)}{x} \;/\; \mu_C(x) = \min \left[\mu_A(x), \mu_{\bar{B}}(x) \right] \right\}$$

Ex: Let $X = \{ a, b, c, d, e, \}$ be the universal set

and $A = \left\{ \frac{0.7}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0.3}{d} \right\}$

$B = \left\{ \frac{0.8}{a} + \frac{1}{c} + \frac{0.4}{d} + \frac{0.2}{e} \right\}$

be two fuzzy sets.

Then find: (i) $\bar{A}$, $\bar{B}$

(ii) $A \cup B$, $A \cap B$,

(iii) $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$ & $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$ } De Morgan's law.

(iv) Show that

- $A \cup \bar{A} \neq X$ [law of Excluded middle]
- & $A \cap \bar{A} \neq \phi$ [law of Contradiction

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(8)

Soln :⇒ Given fuzzy sets are

$$A = \left\{ \frac{0.7}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0.3}{d} + \frac{0}{e} \right\} \quad \text{---(1)}$$

$$B = \left\{ \frac{0.8}{a} + \frac{0}{b} + \frac{1}{c} + \frac{0.4}{d} + \frac{0.2}{e} \right\} \quad \text{---(2)}$$

(i) To find $\overline{A}$ (By using defn of complement of fuzzy set)

$$\overline{A} = \left\{ \frac{1-(0.7)}{a} + \frac{1-(0.4)}{b} + \frac{1-(0.5)}{c} + \frac{1-(0.3)}{d} + \frac{1-0}{e} \right\}$$

$$= \left\{ \frac{0.3}{a} + \frac{0.6}{b} + \frac{0.5}{c} + \frac{0.7}{d} + \frac{1}{e} \right\} \quad \text{---(3)}$$

To find $\overline{B}$

$$\overline{B} = \left\{ \frac{1-(0.8)}{a} + \frac{1-0}{b} + \frac{1-1}{c} + \frac{1-(0.4)}{d} + \frac{1-0.2}{e} \right\}$$

$$= \left\{ \frac{0.2}{a} + \frac{1}{b} + \frac{0}{c} + \frac{0.6}{d} + \frac{0.8}{e} \right\} \quad \text{---(4)}$$

(ii) To find $A \cup B$ [using defn of union of fuzzy set]

$$A \cup B = \left\{ \frac{\max(0.7, 0.8)}{a} + \frac{\max(0.4, 0)}{b} + \frac{\max(0.5, 1)}{c} + \frac{\max(0.3, 0.4)}{d} + \frac{\max(0, 0.2)}{e} \right\}$$

$$A \cup B = \left\{ \frac{0.8}{a} + \frac{0.4}{b} + \frac{1}{c} + \frac{0.4}{d} + \frac{0.2}{e} \right\}$$

---(5)

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To find $A \cap B$ [using defn of Intersection of fuzzy set]

$$A \cap B = \left\{ \frac{\min(0.7, 0.8)}{a} + \frac{\min(0.4, 0)}{b} + \frac{\min(0.5, 1)}{c} + \frac{\min(0.3, 0.4)}{d} + \frac{\min(0, 0.2)}{e} \right\}$$

$$\Rightarrow A \cap B = \left\{ \frac{0.7}{a} + \frac{0}{b} + \frac{0.5}{c} + \frac{0.3}{d} + \frac{0}{e} \right\} \quad \text{---(6)}$$

(iii) To prove De-Morgan's law

Here from eqn (5) & (6) and using defn of complement

$$\overline{A \cup B} = \left\{ \frac{1-0.8}{a} + \frac{1-0.4}{b} + \frac{1-1}{c} + \frac{1-0.4}{d} + \frac{1-0.2}{e} \right\}$$

$$= \left\{ \frac{0.2}{a} + \frac{0.6}{b} + \frac{0}{c} + \frac{0.6}{d} + \frac{0.8}{e} \right\} \quad \text{---(7)}$$

& $$\overline{A \cap B} = \left\{ \frac{1-0.7}{a} + \frac{1-0}{b} + \frac{1-0.5}{c} + \frac{1-0.3}{d} + \frac{1-0}{e} \right\}$$

$$= \left\{ \frac{0.3}{a} + \frac{1}{b} + \frac{0.5}{c} + \frac{0.7}{d} + \frac{1}{e} \right\} \quad \text{---(8)}$$

Now using eqn (3) & (4) and By defn of intersection & union

$$\bar{A} \cap \bar{B} = \left\{ \frac{\min(0.3, 0.2)}{a} + \frac{\min(0.6, 1)}{b} + \frac{\min(0.5, 0)}{c} + \frac{\min(0.7, 0.6)}{d} + \frac{\min(1, 0.8)}{e} \right\}$$

$$= \left\{ \frac{0.2}{a} + \frac{0.6}{b} + \frac{0}{c} + \frac{0.6}{d} + \frac{0.8}{e} \right\} \quad \text{---(9)}$$

and $$\bar{A} \cup \bar{B} = \left\{ \frac{\max(0.3, 0.2)}{a} + \frac{\max(0.6, 1)}{b} + \frac{\max(0.5, 0)}{c} + \frac{\max(0.7, 0.6)}{d} + \frac{\max(1, 0.8)}{e} \right\}$$

$$= \left\{ \frac{0.3}{a} + \frac{1}{b} + \frac{0.5}{c} + \frac{0.7}{d} + \frac{1}{e} \right\} \quad \text{---(10)}$$

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∴ from ⑦ & ⑨ we have

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

and from ⑧ & ⑩ we have

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Hp.

(IV) To show $A \cup \overline{A} \neq X$ [Law of Excluded middle]

from ① & ③ and using defn of union, we get

$$A \cup \overline{A} = \left\{ \frac{\max(0.7, 0.3)}{a} + \frac{\max(0.4, 0.6)}{b} + \frac{\max(0.5, 0.5)}{c} + \frac{\max(0.3, 0.7)}{d} + \frac{\max(0,1)}{e} \right\}$$

$$= \left\{ \frac{0.7}{a} + \frac{0.6}{b} + \frac{0.5}{c} + \frac{0.7}{d} + \frac{1}{e} \right\}$$

⇒ and X is universal set having membership grade of each element is unity

ie. $X = \left\{ \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} \right\}$

∴ Thus $\boxed{A \cup \overline{A} \neq X}$ Hp.

To show $A \cap \overline{A} \neq \phi$ [Law of contradiction]

$$\Rightarrow A \cap \overline{A} = \left\{ \frac{\min(0.7, 0.3)}{a} + \frac{\min(0.4, 0.6)}{b} + \frac{\min(0.5, 0.5)}{c} + \frac{\min(0.3, 0.7)}{d} + \frac{\min(0,1)}{e} \right\}$$

$$= \left\{ \frac{0.3}{a} + \frac{0.4}{b} + \frac{0.5}{c} + \frac{0.3}{d} + \frac{0}{e} \right\}$$

and ϕ is empty fuzzy set having membership grade of all element is zero. ie. $\phi = \left\{ \frac{0}{a}, \frac{0}{b}, \frac{0}{c}, \frac{0}{d}, \frac{0}{e} \right\}$

$\Rightarrow \boxed{A \cap \overline{A} \neq \phi}$ Hp.

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Ex: let $A = \{(x_1, 0.2), (x_2, 0.7), (x_3, 0.9)\}$

$B = \{(x_1, 1), (x_2, 0.4), (x_3, 0.7), (x_4, 0.3)\}$

$C = \{(x_1, 0.1), (x_3, 0.8), (x_4, 0.5)\}$

are any three fuzzy sets. ~~and~~ defined on ~~universal set~~ element of universal set X.

Show that. (i) $\overline{A \cup B} = \overline{A} \cap \overline{B}$

(ii) $\overline{B \cap C} = \overline{B} \cup \overline{C}$

(iii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributing law)

Soln: Here universe of discourse $X = \{x_1, x_2, x_3, x_4\}$

and given fuzzy sets are

$$A = \left\{ \frac{0.2}{x_1} + \frac{0.7}{x_2} + \frac{0.9}{x_3} + \frac{0}{x_4} \right\} \quad \text{---(1)}$$

$$B = \left\{ \frac{1}{x_1} + \frac{0.4}{x_2} + \frac{0.7}{x_3} + \frac{0.3}{x_4} \right\} \quad \text{---(2)}$$

$$C = \left\{ \frac{0.1}{x_1} + \frac{0}{x_2} + \frac{0.8}{x_3} + \frac{0.5}{x_4} \right\} \quad \text{---(3)}$$

∴ By defn of complement

$$\overline{A} = \left\{ \frac{1-0.2}{x_1} + \frac{1-0.7}{x_2} + \frac{1-0.9}{x_3} + \frac{1-0}{x_4} \right\} = \left\{ \frac{0.8}{x_1} + \frac{0.3}{x_2} + \frac{0.1}{x_3} + \frac{1}{x_4} \right\} \quad \text{---(4)}$$

$$\overline{B} = \left\{ \frac{1-1}{x_1} + \frac{1-0.4}{x_2} + \frac{1-0.7}{x_3} + \frac{1-0.3}{x_4} \right\} = \left\{ \frac{0}{x_1} + \frac{0.6}{x_2} + \frac{0.3}{x_3} + \frac{0.7}{x_4} \right\} \quad \text{---(5)}$$

& $\overline{C} = \left\{ \frac{1-0.1}{x_1} + \frac{1-0}{x_2} + \frac{1-0.8}{x_3} + \frac{1-0.5}{x_4} \right\}$

$$= \left\{ \frac{0.9}{x_1} + \frac{1}{x_2} + \frac{0.2}{x_3} + \frac{0.5}{x_4} \right\} \quad \text{---(6)}$$

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(i) To prove (i) $\overline{A \cup B} = \overline{A} \cap \overline{B}$

From (1) & (2) and By using defn of union. we have

$$A \cup B = \left\{ \frac{\max(0.2, 1)}{x_1} + \frac{\max(0.7, 0.4)}{x_2} + \frac{\max(0.9, 0.7)}{x_3} + \frac{\max(0, 0.3)}{x_4} \right\}$$

$$= \left\{ \frac{1}{x_1} + \frac{0.7}{x_2} + \frac{0.9}{x_3} + \frac{0.3}{x_4} \right\} \quad \text{——— (7)}$$

By defn of complement

$$\overline{A \cup B} = \left\{ \frac{1-1}{x_1} + \frac{1-0.7}{x_2} + \frac{1-0.9}{x_3} + \frac{1-0.3}{x_4} \right\}$$

$$= \left\{ \frac{0}{x_1} + \frac{0.3}{x_2} + \frac{0.1}{x_3} + \frac{0.7}{x_4} \right\} \quad \text{——— (8)}$$

also. using (4) & (5) and defn of intersection

$$\overline{A} \cap \overline{B} = \left\{ \frac{\min(0.8, 0)}{x_1} + \frac{\min(0.3, 0.6)}{x_2} + \frac{\min(0.1, 0.3)}{x_3} + \frac{\min(1, 0.7)}{x_4} \right\}$$

$$= \left\{ \frac{0}{x_1} + \frac{0.3}{x_2} + \frac{0.1}{x_3} + \frac{0.7}{x_4} \right\} \quad \text{——— (9)}$$

from (8) & (9) we have

$$\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$$ H.P.

(ii) To prove $\overline{B \cap C} = \overline{B} \cup \overline{C}$

Form (2) & (3) and using defn of intersection

$$B \cap C = \left\{ \frac{\min(1, 0.1)}{x_1} + \frac{\min(0.4, 0)}{x_2} + \frac{\min(0.7, 0.8)}{x_3} + \frac{\min(0.3, 0.5)}{x_4} \right\}$$

$$= \left\{ \frac{0.1}{x_1} + \frac{0}{x_2} + \frac{0.7}{x_3} + \frac{0.3}{x_4} \right\} \quad \text{——— (10)}$$

By using defn of complement

$$\overline{B \cap C} = \left\{ \frac{1-0.1}{x_1} + \frac{1-0}{x_2} + \frac{1-0.7}{x_3} + \frac{1-0.3}{x_4} \right\}$$

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(13)

$$\overline{B \cap C} = \left\{ \frac{0.9}{x_1} + \frac{1}{x_2} + \frac{0.3}{x_3} + \frac{0.7}{x_4} \right\} \quad \text{---(11)} \quad (7)$$

Now, from (5) & (6) and by defn of union

$$\bar{B} \cup \bar{C} = \left\{ \frac{\max(0, 0.9)}{x_1} + \frac{\max(0.6, 1)}{x_2} + \frac{\max[0.3, 0.2)}{x_3} + \frac{\max(0.7, 0.5)}{x_4} \right\}$$

$$= \left\{ \frac{0.9}{x_1} + \frac{1}{x_2} + \frac{0.3}{x_3} + \frac{0.7}{x_4} \right\} \quad \text{---(12)}$$

Hence from (11) & (12) we have

$$\boxed{\overline{B \cap C} = \bar{B} \cup \bar{C}}$$ H.P.

(11) To prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

using eqn (1) & (10) and By defn of union

$$A \cup (B \cap C) = \left\{ \frac{\max(0.2, 0.1)}{x_1} + \frac{\max(0.7, 0)}{x_2} + \frac{\max(0.9, 0.7)}{x_3} + \frac{\max(0, 0.3)}{x_4} \right\}$$

$$= \left\{ \frac{0.2}{x_1} + \frac{0.7}{x_2} + \frac{0.9}{x_3} + \frac{0.3}{x_4} \right\} \quad \text{---(13)}$$

Now, using (1) & (3) and by defn of union.

$$A \cup C = \left\{ \frac{\max(0.2, 0.1)}{x_1} + \frac{\max(0.7, 0)}{x_2} + \frac{\max(0.9, 0.8)}{x_3} + \frac{\max(0, 0.5)}{x_4} \right\}$$

$$= \left\{ \frac{0.2}{x_1} + \frac{0.7}{x_2} + \frac{0.9}{x_3} + \frac{0.5}{x_4} \right\} \quad \text{---(14)}$$

Now, using (7) & (14) and By defn of intersection

$$(A \cup B) \cap (A \cup C) = \left\{ \frac{\min(1, 0.2)}{x_1} + \frac{\min(0.7, 0.7)}{x_2} + \frac{\min(0.9, 0.9)}{x_3} + \frac{\min(0.3, 0.5)}{x_4} \right\}$$

$$= \left\{ \frac{0.2}{x_1} + \frac{0.7}{x_2} + \frac{0.9}{x_3} + \frac{0.3}{x_4} \right\} \quad \text{---(15)}$$

From (14) & (15)

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$$

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Ex Let $X = \{0, 1, 2, 3, 4, 5\}$ be the universal set and A, B & C are any three fuzzy set having membership function $\mu_A(x) = \frac{x}{x+2}$, $\mu_B(x) = 2^{-x}$

$\mu_C(x) = \frac{1}{1+10(x-2)^2}$ define on all $x \in X$. respectively.

Then find (i) fuzzy set A, B and C

(ii) show that $A \cup (\bar{A} \cap B) = A \cup B$

Soln :⇒ let $X = \{0, 1, 2, 3, 4, 5\}$ be the universal set

let A be the fuzzy set defined as

$$A = \left\{ \frac{\mu_A(x)}{x} \;/\; \text{where } \mu_A(x) = \frac{x}{x+2} \text{ for all } x \in X \right\}$$

$$\therefore A = \left\{ \frac{0}{0} + \frac{0.33}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.67}{4} + \frac{0.71}{5} \right\} \quad \text{---(1)}$$

Let B be the fuzzy set given by

$$B = \left\{ \frac{\mu_B(x)}{x} \;/\; \mu_B(x) = 2^{-x} = \frac{1}{2^x}, \forall x \in X \right\}$$

$$\therefore B = \left\{ \frac{1}{0} + \frac{0.5}{1} + \frac{0.25}{2} + \frac{0.125}{3} + \frac{0.0625}{4} + \frac{0.0315}{5} \right\} \quad \text{---(2)}$$

and let C be the fuzzy set given by

$$C = \left\{ \frac{\mu_C(x)}{x} \;/\; \mu_C(x) = \frac{1}{1+10(x-2)^2} \quad \forall x \in X \right\}$$

$$\therefore C = \left\{ \frac{0.024}{0} + \frac{0.09}{1} + \frac{1}{2} + \frac{0.09}{3} + \frac{0.024}{4} + \frac{0.01}{5} \right\} \quad \text{---(3)}$$

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(ii) Now To prove $A \cup (\bar{A} \cap B) = A \cup B$

from eq^n (1) and By defn of complement

$$\bar{A} = \left\{ \frac{1-0}{0} + \frac{1-0.33}{1} + \frac{1-0.5}{2} + \frac{1-0.6}{3} + \frac{1-0.67}{4} + \frac{1-0.71}{5} \right\}$$

$$= \left\{ \frac{1}{0} + \frac{0.67}{1} + \frac{0.5}{2} + \frac{0.4}{3} + \frac{0.33}{4} + \frac{0.29}{5} \right\} \quad \text{---(5)}$$

Now using (5) and (2) and by defn of intersection

$$\bar{A} \cap B = \left\{ \frac{\min(1,1)}{0} + \frac{\min(0.67,0.5)}{1} + \frac{\min(0.5,0.25)}{2} + \frac{\min(0.4,0.125)}{3} + \frac{\min(0.33,0.0625)}{4} + \frac{\min(0.29,0.0315)}{5} \right\}$$

$$= \left\{ \frac{1}{0} + \frac{0.5}{1} + \frac{0.25}{2} + \frac{0.125}{3} + \frac{0.0625}{4} + \frac{0.0315}{5} \right\} \quad \text{---(6)}$$

Hence using (1) & (6) and defn of union. we get

$$A \cup (\bar{A} \cap B) = \left\{ \frac{\max(0,1)}{0} + \frac{\max(0.33,0.5)}{1} + \frac{\max(0.5,0.25)}{2} + \frac{\max(0.6,0.125)}{3} + \frac{\max(0.67,0.0625)}{4} + \frac{\max(0.71,0.0315)}{5} \right\}$$

$$= \left\{ \frac{1}{0} + \frac{0.5}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.67}{4} + \frac{0.71}{5} \right\} \quad \text{---(7)}$$

Now using (1) & (2) and defn of union.

$$A \cup B = \left\{ \frac{\max(0,1)}{0} + \frac{\max(0.33,0.5)}{1} + \frac{\max(0.5,0.25)}{2} + \frac{\max(0.6,0.125)}{3} + \frac{\max(0.67,0.0625)}{4} + \frac{\max(0.71,0.0315)}{5} \right\}$$

$$\Rightarrow A \cup B = \left\{ \frac{1}{0} + \frac{0.5}{1} + \frac{0.5}{2} + \frac{0.6}{3} + \frac{0.67}{4} + \frac{0.71}{5} \right\} \quad \text{---(8)}$$

Hence from (7) & (8)

$$\boxed{A \cup (\bar{A} \cap B) = A \cup B}$$

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★ Fuzzy Relation

① Cartesian product of Fuzzy sets

Let x and Y be the two universe of discourse (some or distinct). and A & B are two distinct fuzzy set defined on elements $x \in X$ and $y \in Y$ respectively having membership function $\mu_A(x)$ & $\mu_B(y)$. respectively.

Then the cartesian product of two fuzzy set A & B is given by

$$R = A \times B = \left\{ \frac{\mu_R(x,y)}{(x,y)} \;/\; \mu_R(x,y) = \min\left[\mu_A(x), \mu_B(y)\right] \right.$$
$$\left. \text{for all } x \in X, \text{ \& } y \in Y \right\}$$

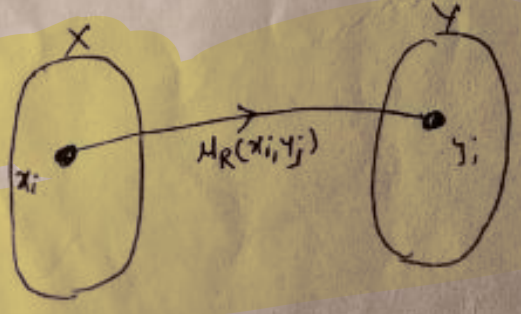
Matrix Representation :

The cartesian product fuzzy set $A \times B$ can be represented by in matrix form as follows

$$M_R = \begin{matrix} \\ x_i \end{matrix} \overset{\begin{matrix} y_j \end{matrix}}{\left[\mu_R(x_i, y_j) \right]_{n \times m}}$$

Diagram Representation

We can be also represented by arrow diagram / diagraph by assigning the weight of edge [$\mu_R(x_i, y_j)$ for edge betn $x_i \in x$ and $y_j \in Y$.]



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Ex: Let $X=\{a, b, c\}$ & $Y=\{x, y, z\}$ are two universal set. and A & B are any two fuzzy set defined as.

$$A=\left\{\frac{0.7}{a}+\frac{0.5}{b}+\frac{0.2}{c}\right\}$$

$$B=\left\{\frac{0.4}{x}+\frac{0.6}{y}\right\}$$

Then find ① $A \times B$

Hence Represent by matrix and diagraph.

Soln: Let $X=\{a, b, c\}$ and $Y=\{x, y, z\}$ be the two universal set.

and $A=\left\{\frac{0.7}{a}+\frac{0.5}{b}+\frac{0.2}{c}\right\}$

$B=\left\{\frac{0.4}{x}+\frac{0.6}{y}+\frac{0}{z}\right\}$ be two fuzzy set

$$\therefore A \times B=\left\{\frac{\min[0.7,0.4]}{(a,x)}+\frac{\min[0.7,0.6]}{(a,y)}+\frac{\min[0.7,0]}{(a,z)}\right.$$

$$+\frac{\min[0.5,0.4]}{(b,x)}+\frac{\min[0.5,0.6]}{(b,y)}+\frac{\min[0.5,0]}{(b,z)}$$

$$\left.+\frac{\min[0.2,0.4]}{(c,x)}+\frac{\min[0.2,0.6]}{(c,y)}+\frac{\min[0.2,0]}{(c,z)}\right\}$$

$$\Rightarrow A \times B=\left\{\frac{0.4}{(a,x)}+\frac{0.6}{(a,y)}+\frac{0}{(a,z)}+\frac{0.4}{(b,x)}+\frac{0.5}{(b,y)}+\right.$$

$$\left.\frac{0}{(b,z)}+\frac{0.2}{(c,x)}+\frac{0.2}{(c,y)}+\frac{0}{(c,z)}\right\}$$

Matrix Representation

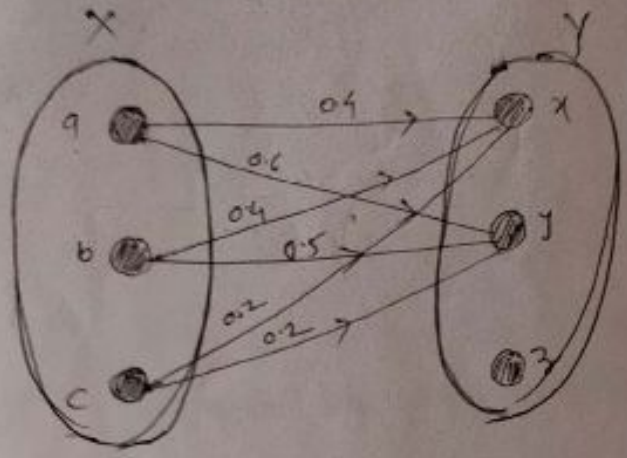
M_R =

	x	y	z
a	0.4	0.6	0
b	0.4	0.5	0
c	0.2	0.2	0

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Digraph Representation (Sagittal Diagraph)



Ex: Let $X = \{1, 2, 3, 4\}$ be the universal set and $A = \left\{\frac{0.3}{1}, \frac{0.5}{2}, \frac{0.8}{4}\right\}$, $B = \left\{\frac{0.7}{1}, \frac{0.9}{3}, \frac{0.4}{4}\right\}$ are any two fuzzy set defined on element of X.

Then Find (i) $R = A \times B$

(ii) Matrix M_R (iii) Diagraph For R.

Soln. Let $A = \left\{\frac{0.3}{1} + \frac{0.5}{2} + \frac{0}{3} + \frac{0.8}{4}\right\}$

$B = \left\{\frac{0.7}{1} + \frac{0}{2} + \frac{0.9}{3} + \frac{0.4}{4}\right\}$

are any two fuzzy set defined on $X = \{1, 2, 3, 4\}$

$$\therefore A \times B = \left\{\frac{\min(0.3, 0.7)}{(1,1)} + \frac{\min(0.3, 0)}{(1,2)} + \frac{\min(0.3, 0.9)}{(1,3)} + \frac{\min(0.3, 0.4)}{(1,4)}\right.$$

$$+ \frac{\min(0.5, 0.7)}{(2,1)} + \frac{\min(0.5, 0)}{(2,2)} + \frac{\min(0.5, 0.9)}{(2,3)} + \frac{\min(0.5, 0.4)}{(2,4)}$$

$$+ \frac{\min(0, 0.7)}{(3,1)} + \frac{\min(0, 0)}{(3,2)} + \frac{\min(0, 0.9)}{(3,3)} + \frac{\min(0, 0.4)}{(3,4)}$$

$$\left. \frac{\min(0.8, 0.7)}{(4,1)} + \frac{\min(0.8, 0)}{(4,2)} + \frac{\min(0.8, 0.9)}{(4,3)} + \frac{\min(0.8, 0.4)}{(4,4)}\right\}$$

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$A \times B = \{ \frac{0.3}{(1,1)} + \frac{0}{(1,2)} + \frac{0.3}{(1,3)} + \frac{0.3}{(1,4)} + \frac{0.5}{(2,1)} + \frac{0}{(2,2)} + \frac{0.5}{(2,3)} + \frac{0.4}{(2,4)} + \frac{0}{(3,1)} + \frac{0}{(3,2)} + \frac{0}{(3,3)} + \frac{0}{(3,4)} + \frac{0.7}{(4,1)} + \frac{0}{(4,2)} + \frac{0.8}{(4,3)} + \frac{0.4}{(4,4)} \}$

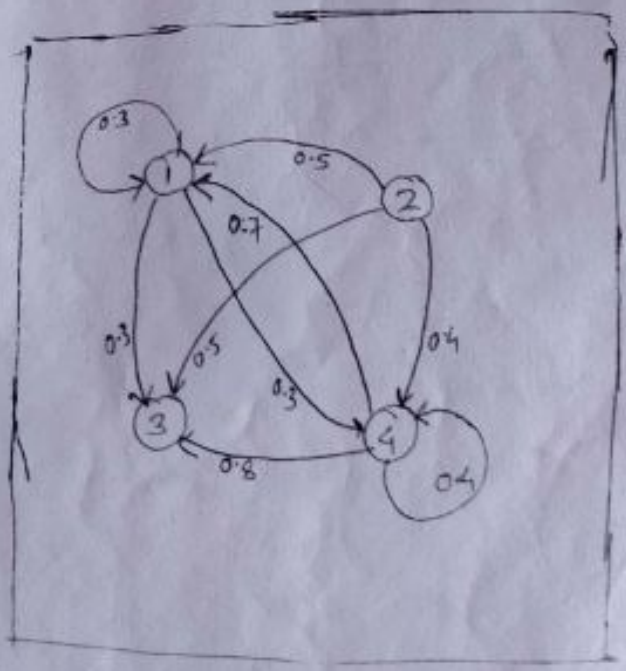
$R = A \times B = \{ \frac{0.3}{(1,1)} + \frac{0.3}{(1,3)} + \frac{0.3}{(1,4)} + \frac{0.5}{(2,1)} + \frac{0.5}{(2,3)} + \frac{0.4}{(2,4)} + \frac{0.7}{(4,1)} + \frac{0.8}{(4,3)} + \frac{0.4}{(4,4)} \}$

Matrix Representation

$M_R =$

	1	2	3	4
1	0.3	0	0.3	0.3
2	0.5	0	0.5	0.4
3	0	0	0	0
4	0.7	0	0.8	0.4

Digraph



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Composition (Composite Relation) (max-min product)

Let X, Y, Z be three universal sets and A, B & C are any three fuzzy set defined on element of universal sets respective with membership grade $\mu_A(x)$, $\mu_B(y)$, $\mu_B(z)$ for all $x \in X$, $y \in Y$, & $z \in Z$.

Let $R = A \times B \subseteq X \times Y$ & $S = B \times C \subseteq Y \times Z$ be the any two relation (cartesian product).

Then the composite relation $T = R \circ S$ is relation from set A to C defined as

$$T = R \circ S = \left\{ \frac{\mu_T(x,z)}{(x,z)} \;/\; \mu_T(x,z) = \max\left[\min\left(\mu_R(x,y), \mu_S(y,z)\right)\right] \right\}$$

$$\subseteq X \times Z$$

Ex: ⇒ let $X = \{a, b, c\}$, $Y = \{x, y, z, w\}$, $Z = \{1, 2, 3\}$ be the theree universal set and

$$R = A \times B = \left\{ \frac{0.7}{(a,x)}, \frac{0.3}{(a,z)}, \frac{0.4}{(b,y)}, \frac{0.7}{(b,z)}, \frac{1}{(c,z)}, \frac{0.2}{(c,w)} \right\}$$

$$S = B \times C = \left\{ \frac{1}{(x,1)}, \frac{0.3}{(x,3)}, \frac{0.1}{(y,1)}, \frac{0.2}{(y,2)}, \frac{0.5}{(y,3)}, \frac{0.9}{(z,3)}, \frac{0.8}{(w,1)}, \frac{1}{(w,2)} \right\}$$

Find (i) M_R, M_S. (ii) Diagraph of R & S

(ii) composite relation $T = R \circ S$

also find M_T and Diagraph T.

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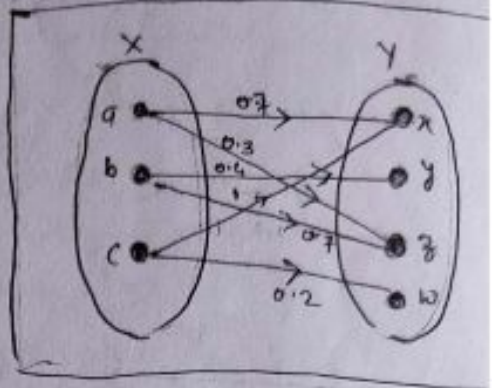
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Sol: Given that

$$R = A \times B = \left\{ \frac{0.7}{(a,x)} + \frac{0.3}{(a,z)} + \frac{0.4}{(b,y)} + \frac{0.7}{(b,z)} + \frac{1}{(c,x)} + \frac{0.2}{(c,w)} \right\} \subseteq X \times Y$$

$M_R =$

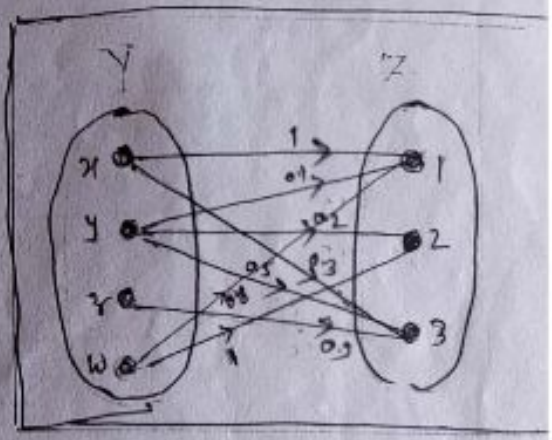
	x	y	z	w
a	0.7	0	0.3	0
b	0	0.4	0.7	0
c	1	0	0	0.2



and $S = B \times C = \left\{ \frac{1}{(x,1)} + \frac{0.3}{(x,3)} + \frac{0.1}{(y,1)} + \frac{0.2}{(y,2)} + \frac{0.5}{(y,3)} + \frac{0.9}{(z,3)} + \frac{0.8}{(w,1)} + \frac{1}{(w,2)} \right\} \subseteq Y \times Z$

$M_S =$

	1	2	3
x	1	0	0.3
y	0.1	0.2	0.5
z	0	0	0.9
w	0.8	1	0



Now To find $T = R \circ S$

$R \circ S = T = A \times C \subseteq X \times Z$

$= \left\{ \frac{\mu_T(\alpha,\beta)}{(\alpha,\beta)} \right.$ / $\mu_T = \max \left[\min \left(\mu_R(x,y), \mu_S(y,z) \right) \right] \Big\}$

where

$\alpha \in A \subseteq X$

$\gamma \in B \subseteq Y$

$\beta \in C \subseteq Z$

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$\mu_T(a,1) = \max[\min(0.7,1), \min(0,0.1), \min(0.3,0), \min(0,0.8)]$

$= \max.[\ 0.7\ ,\ 0,\ 0,\ 0\ ] = 0.7$

$\mu_T(a,2) = \max[\min(0.7,0), \min(0,0.2), \min(0.3,0), \min(0,1)]$

$= \max.[\ 0,\ 0,0,0\ ] = 0$

$\mu_T(a,3) = \max[\min(0.7,0.3), \min(0,0.5), \min(0.3,0.9), \min(0,0)]$

$= \max[0.3,\ 0,\ 0.3,\ 0\ ] = 0.3.$

$\mu_T(b,1) = \max.[\min(0,1), \min(0.4,0.1), \min(0.7,0), \min(0,0.8)]$

$= \max[\ 0,\ 0.1,\ 0,0\ ] = 0.1$

$\mu_T(b,2) = \max[\min(0,0), \min(0.4,0.2), \min(0.7,0), \min(0,0.1)$

$= \max.[0,\ 0.2,\ 0,\ 0\ ] = 0.2$

$\mu_T(b,3) = \max[\min(0,0.3), \min(0.4,0.5), \min(0.7,0.9), \min(0,0)]$

$= \max[0,\ 0.4,\ 0.7,\ 0\ ] = 0.7.$

$\mu_T(c,1) = \max.[\min(1,1), \min(0,0.1), \min(0,0), \min(0.2,0.8)]$

$= \max[1,\ 0,\ 0,\ 0.2\ ] = 1.$

$\mu_T(c,2) = \max.[\min(1,0), \min(0,0.2), \min(0,0), \min(0.2,1)$

$= \max[0,\ 0,\ 0,\ 0.2\ ] = 0.2$

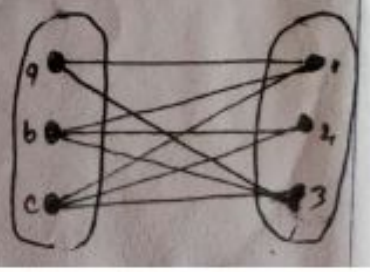
$\mu_T(c,3) = \max[\min(1,0.3), \min(0,0.5), \min(0,0.9), \min(0.2,0)$

$= \max[0.3,\ 0,\ 0,\ 0\ ] = 0.3$

$$R_{OS} = T = \left\{ \frac{0.7}{(a,1)} + \frac{0}{(a,2)} + \frac{0.3}{(a,3)} + \frac{0.1}{(b,1)} + \frac{0.2}{(b,2)} + \frac{0.7}{(b,3)} + \frac{1}{(c,1)} + \frac{0.2}{(c,2)} + \frac{0.3}{(c,3)} \right\}$$

$M_T =$

	1	2	3
a	0.7	0	0.3
b	0.1	0.2	0.7
c	1	0.2	0.3



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Algebraic product

Let A & B are any two fuzzy set defined on element of X & Y (universal sets) with membership grade $\mu_A(x)$ for $x \in X$ & $\mu_B(y)$ for $y \in Y$. respectively

Then their algebraic product is defined and denoted as

$$C = A \cdot B = \left\{ \frac{\mu_C(x,y)}{(x,y)} \;/\; \mu_C = \mu_A(x) \cdot \mu_B(y) \right\}$$

for all $x \in X$, $y \in Y$.

Concentration of fuzzy set

Let A be a fuzzy set defined for all $x \in X$, with membership grade $\mu_A(x) \in [0,1]$. then the concentration of A is denoted as

$$\text{Con}(A) = A^2 = A \cdot A = \left\{ \frac{\mu_{A^2}(x)}{x} \;/\; \mu_{A^2} = (\mu_A(x))^2 \right\}$$

For eg: let $A = \left\{ \frac{1}{0} + \frac{0.7}{1} + \frac{0.3}{2} + \frac{0.5}{(0.3)} \right\}$

Then $\text{Con}(A) = A \cdot A$

$$= \left\{ \frac{(1\times1)}{0} + \frac{(0.7\times0.7)}{1} + \frac{(0.3)\times(0.3)}{2} + \frac{(0.5)(0.5)}{3} \right]$$

$$= \left\{ \frac{1}{0} + \frac{0.49}{1} + \frac{0.09}{2} + \frac{0.25}{3} \right\}$$

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Dilation of fuzzy set:

Let A be the fuzzy set defined for all $x \in X$ having membership grade $\mu_A(x) \in [0,1]$. Then it's Dilation is given by

$$DIL(A) = \left\{ \frac{\sqrt{\mu_A(x)}}{x} \;/\; \mu_A(x) \in [0,1], \text{ for all } x \in A \subseteq X \right\}$$

For eg. Let $A = \left\{ \frac{0.49}{a} + \frac{0.25}{b} + \frac{0.36}{c} \right\}$

$$\therefore DIL(A) = \left\{ \frac{\sqrt{0.49}}{a} + \frac{\sqrt{0.25}}{b} + \frac{\sqrt{0.36}}{c} \right\}$$

$$= \left\{ \frac{0.7}{a} + \frac{0.5}{b} + \frac{0.6}{c} \right\}$$

Note: $CON(DIL(A)) \neq A$

$\neq DIL(CON(A))$

Ex: Let $A = \left\{ \frac{0.1}{\alpha} + \frac{0.3}{\beta} + \frac{0.4}{\gamma} + \frac{0.6}{\theta} \right\} \subseteq X$

$$B = \left\{ \frac{0.7}{a} + \frac{0.6}{b} + \frac{0.9}{c} \right\} \subseteq Y$$

Then Find (i) $A \cdot B$, $A \times B$.

(ii) $CON(A)$, $CON(B)$.

(iii) $DIL(A)$ $DIL(B)$.

Soln: Let given fuzzy sets are

$$A = \left\{ \frac{0.1}{\alpha} + \frac{0.3}{\beta} + \frac{0.4}{\gamma} + \frac{0.6}{\theta} \right\}$$

& $$B = \left\{ \frac{0.7}{a} + \frac{0.6}{b} + \frac{0.9}{c} \right\}$$

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$$A \cdot B = \left\{ \frac{(0.1)(0.7)}{(\alpha,a)} + \frac{(0.1)(0.6)}{(\alpha,b)} + \frac{(0.1)\cdot(0.9)}{(\alpha,c)} + \frac{(0.3)(0.7)}{(\beta,a)} \right.$$
$$+ \frac{(0.3)(0.6)}{(\beta,b)} + \frac{(0.3)(0.9)}{(\beta,c)} + \frac{(0.4)(0.7)}{(\gamma,a)} + \frac{(0.4)(0.6)}{(\gamma,b)}$$
$$\left. + \frac{(0.4)(0.9)}{(\gamma,c)} + \frac{(0.6)(0.7)}{(\theta,a)} + \frac{(0.6)(0.6)}{(\theta,b)} + \frac{(0.6)(0.9)}{(\theta,c)} \right\}$$

$$= \left\{ \frac{0.07}{(\alpha,a)} + \frac{0.06}{(\alpha,b)} + \frac{0.09}{(\alpha,c)} + \frac{0.21}{(\beta,a)} + \frac{0.18}{(\beta,b)} \right.$$
$$+ \frac{0.27}{(\beta,c)} + \frac{0.28}{(\gamma,a)} + \frac{0.24}{(\gamma,b)} + \frac{0.36}{(\gamma,c)} + \frac{0.42}{(\theta,a)}$$
$$\left. + \frac{0.36}{(\theta,b)} + \frac{0.54}{(\theta,c)} \right\}$$

NOW

$$A \times B = \left\{ \frac{\min(0.1, 0.7)}{(\alpha,a)} + \frac{\min(0.1, 0.6)}{(\alpha,b)} + \frac{\min(0.1, 0.9)}{(\alpha,c)} \right.$$
$$+ \frac{\min(0.3, 0.7)}{(\beta,a)} + \frac{\min(0.3, 0.6)}{(\beta,b)} + \frac{\min(0.3, 0.9)}{(\beta,c)}$$
$$+ \frac{\min(0.4, 0.7)}{(\gamma,a)} + \frac{\min(0.4, 0.6)}{(\gamma,b)} + \frac{\min(0.4, 0.9)}{(\gamma,c)}$$
$$\left. + \frac{\min(0.6, 0.7)}{(\theta,a)} + \frac{\min(0.6, 0.6)}{(\theta,b)} + \frac{\min(0.6, 0.9)}{(\theta,c)} \right\}$$

$$= \left\{ \frac{0.1}{(\alpha,a)} + \frac{0.1}{(\alpha,b)} + \frac{0.1}{(\alpha,c)} + \frac{0.3}{(\beta,a)} + \frac{0.3}{(\beta,b)} \right.$$
$$\frac{0.3}{(\beta,c)} + \frac{0.4}{(\gamma,a)} + \frac{0.4}{(\gamma,b)} + \frac{0.4}{(\gamma,c)} + \frac{0.6}{(\theta,a)}$$
$$\left. + \frac{0.6}{(\theta,b)} + \frac{0.6}{(\theta,c)} \right\}$$

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(ii) To find Con (A) & Con. (B).

$$Con(A) = \left\{ \frac{(0.1)^2}{\alpha} + \frac{(0.3)^2}{\beta} + \frac{(0.4)^2}{\gamma} + \frac{(0.6)^2}{\theta} \right\}$$

$$= \left\{ \frac{0.01}{\alpha} + \frac{0.09}{\beta} + \frac{0.16}{\gamma} + \frac{0.36}{\theta} \right\}$$

$$Con(B) = \left\{ \frac{(0.7)^2}{a} + \frac{(0.6)^2}{b} + \frac{(0.9)^2}{c} \right\}$$

$$= \left\{ \frac{0.49}{a} + \frac{0.36}{b} + \frac{0.81}{c} \right\}$$

(iii) Dilation of A & B

$$DIL(A) = \left\{ \frac{\sqrt{0.1}}{\alpha} + \frac{\sqrt{0.3}}{\beta} + \frac{\sqrt{0.4}}{c} + \frac{\sqrt{0.6}}{\theta} \right\}$$

$$= \left\{ \frac{0.32}{\alpha} + \frac{0.55}{\beta} + \frac{0.63}{c} + \frac{0.77}{\theta} \right\}$$

$$DIL(B) = \left\{ \frac{\sqrt{0.7}}{a} + \frac{\sqrt{0.6}}{b} + \frac{\sqrt{0.9}}{c} \right\}$$

$$= \left\{ \frac{0.84}{a} + \frac{0.77}{b} + \frac{0.95}{c} \right\}$$

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Fuzzy Logic :⇒

In a classical logic, A proposition (or statement) have clearly defined boundaries. i.e. the proposition will have truth value 1 (if True) or truth value 0 (if false).

For eg: P: John is six feet tall.

Then. Truth value of P is.

$$T(P) = \begin{cases} 0, & \text{if } P \text{ is false} \\ 1, & \text{if } P \text{ is true.} \end{cases}$$

But some Linguistic proposition such as.

"John is very tall", "It is sunny", "A car is too expensive" etc. are statement does not have clearly defined boundaries, i.e. these statement may be partialy true or partially false. Hence the truth value of such statement will lies bet the value 0, & 1.

ie. IF a proposition with unclearly defined boundry (fuzzy proposition).

Then Truth value of P.

i.e. $T(P) \in [0,1]$

which is similar as the membership function defined in fuzzy set theory.

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~~Logical~~

For eg: Consider a proposition.

P: John is very tall.

and let $A = \{ \frac{0.7}{John}, \frac{0.8}{Jack}, \frac{0.9}{Alice}, \frac{0.4}{Ricky}, \}$

be the fuzzy set of 'tall' pepople.

Thus. $T(P) = 0.7 = \mu_A(John)$.

Logical connectives

In a compound statement, two or more proposition are connected using the connectives "not", "and", "or", "if ... then" etc. are called logical connectives.

(#) Consider any two proposition as follow

P: - x belongs to A $\Rightarrow T(P) = \mu_A(x)$

Q: x belongs to B $\Rightarrow T(Q) = \mu_B(x)$

where A & B are any two fuzzy set

Then

(1) Negation (Not) (symbol. $\exists$ or $\sim$)

$\neg(P)$: x is not belongs To A

$\therefore \boxed{T(\neg P) = 1 - T(P)} = 1 - \mu_A(x)$.

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(ii) Disjunction (OR) : [Symbol 'V']

$P \vee Q$: x belongs to A or x belong to B

$\Rightarrow x \in A$ or $x \in B$

$\Rightarrow x \in (A \cup B)$

∴ Truth value of $P \vee Q$ is given by

$$T(P \vee Q) = \mu_{A \cup B}(x) = \max. [\mu_A(x), \mu_B(x)]$$

$$\Rightarrow \boxed{T(P \vee Q) = \max. [T(P), T(Q)]}$$

(iii) Conjuction (AND) : [Symbol '$\wedge$']

$P \wedge Q$: x belongs to A and x belongs to B.

$\Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in A \cap B$

∴ Thus. $T(P \wedge Q) = \mu_{A \cap B}(x) = \min. [\mu_A(x), \mu_B(x)]$

$$\Rightarrow \boxed{T(P \wedge Q) = \min [T(P), T(Q)]}$$

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(iv) Conditional (If...then) : [symbol '$\Rightarrow$']

$P \rightarrow Q$: If $x \in A$ then $x \in B$

$= x \in [A^c \cup B]$

Thus $T(P \rightarrow Q) = \max . \mu_{\bar{A} \cup B}(x)$

$= \max . [\mu_{\bar{A}}(x), \mu_B(x)]$

$= \max . [(1 - \mu_A(x)), \mu_B(x)]$

$\Rightarrow \boxed{T(P \rightarrow Q) = \max . [1 - T(P)), T(Q)]}$

(v) Biconditional (if and only if) [symbol "$\Leftrightarrow$']

$P \Leftrightarrow Q$: x belongs to A iff x belongs to B

: x (if $x \in A$ then $x \in B$) and (if $x \in B$ then $x \in A$)

: $x \in (\bar{A} \cup B)$ and $x \in (\bar{B} \cup A)$

$\therefore x \in [(\bar{A} \cup B) \cap (\bar{B} \cup A)]$

$\therefore T(P \leftrightarrow Q) = \min [\mu_{\bar{A} \cup B}(x), \min \mu_{\bar{B} \cup A}(x)]$

$= \min [\max(1 - T(P), T(Q)), \max(1 - T(Q), T(P))]$

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Ex: If the truth value of the statement
" John is brave " is 0.7.
and " Jack is brave " is 0.8
Then what is the truth value of the following statement?
" Neither John nor Jack is brave "

Soln: let A be the Fuzzy set of "Brave people"

Since It is given that,

Truth value of statement

P: John is brave ~~is 0.7~~ ⇒ i.e. $T(P) = 0.7$

& Q: Jack is brave i.e. ⇒ $T(Q) = 0.8$

Thus: By using Negation.

$\neg P$: John is not brave ⇒ $T(\neg P) = 1 - T(P)$
$= 1 - 0.7$
$= 0.3$

& $\neg Q$: Jack is not brave ⇒ $T(\neg Q) = 1 - T(Q)$
$= 1 - 0.8$
$= 0.2$

Now given statement is

' Niether John nor Jack is brave '

is equivalent to

John is not brave and Jack is not brave

$\Rightarrow (\neg P \wedge \neg Q)$

Thus

$T(\neg P \wedge \neg Q) = \min [T(\neg P), T(\neg Q)]$

$= \min [0.3, 0.2]$

$= 0.2$

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Let A be the fuzzy set of intelligent peopels.

given by $A = \left\{ \frac{0.7}{\text{William}} + \frac{0.9}{\text{John}} + \frac{0.6}{\text{Alice}} + \frac{0.5}{\text{Ricky}} \right\}$

Find the truth value of the following statements

(i) "William and Ricky both are intelligent"

(ii) Either John or Ricky are intelligent"

(iii) If Alice is intelligent then Ricky is not intelligent"

(iv) "Niether william nor Alice are intelligent."

(v)

: let given fuzzy set of intelligent people is

$$A = \left\{ \frac{0.7}{\text{William}} + \frac{0.9}{\text{John}} + \frac{0.6}{\text{Alice}} + \frac{0.5}{\text{Ricky}} \right\}$$

let P: Williams is intelligent $\Rightarrow T(P) = 0.7$

Q: John is intelligent $\Rightarrow T(Q) = 0.9$

R: Alice is intelligent $\Rightarrow T(R) = 0.6$

S: Ricky is intelligent $\Rightarrow T(S) = 0.5$

(i) To Find Truth value of

"William and Ricky both are Intelligent"

This proposition is equivalent to

$P \wedge S$: william is intelligent and Ricky is intelligent

$\therefore T(P \wedge S) = \min[T(P), T(S)]$

$= \min[0.7, 0.5] = 0.5$

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(ii) To find truth value of

" Either John or Ricky are intelligent"

⇒ QVS : John is intelligent OR Ricky is intelligent

∴ T(QVS) = max. [T(Q), T(S)]

= max [0.9, 0.5]

= 0.9

(iii) To find truth value of

" If Alice is intelligent then Ricky is not intelligent"

This is in symbolic form

⇒ $R \rightarrow \neg S$:

∴ $T(R \rightarrow \neg S)$ = ~~T(¬(R)∧~~

= max. [(1 − T(R)), T(¬S)]

= max [1 − T(R), 1 − T(S)]

= max [(1 − 0.6), (1 − 0.5)]

= max [0.4, 0.5] = 0.5

(iv) To find Truth value.

"Niether William nor Alice are intelligent"

⇒ Williams is not intelligent and Alice is not intelligent

⇒ $(\neg P) \wedge (\neg R)$

∴ $T(\neg P \wedge \neg R)$ = min [T(¬P), T(¬R)]

= min [(1 − T(P)), (1 − T(R))]

= min [(1 − 0.7), (1 − 0.6)] = 0.3

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